

FAHEEM NAZIR

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• [Portfolio](#)

SUMMARY

B.Tech Computer Science student building applied machine-learning systems in Python, with hands-on experience in biomedical signal processing, feature engineering, and rigorous cross-subject model validation. Also fluent in full-stack and backend engineering across React, Node.js, and REST APIs, with a growing focus on reliable and explainable AI.

PROJECTS

PhysioStress AI | Multimodal Physiological Stress Detection (WESAD) | [GitHub Repo](#)

Tech: Python, scikit-learn, NumPy, Pandas, SciPy, Plotly, Streamlit, Signal Processing, Explainable AI

- Built a subject-independent physiological stress classification system using multimodal ECG, EDA, and respiration signals from the WESAD dataset, processing 859 sixty-second windows across 15 subjects at 700 Hz.
- Engineered 28 physiological biomarkers via a zero-phase Butterworth filtering pipeline, R-peak detection and HRV analysis, EDA tonic/phasic decomposition with SCR detection, and respiration cycle analysis.
- Benchmarked four machine-learning models under 15-fold Leave-One-Subject-Out validation with strictly subject-isolated scaling, achieving 94.25% pooled Macro F1 and 0.983 ROC-AUC with Logistic Regression; verified the pipeline against a formal data-leakage audit.
- Developed an interactive research dashboard for signal visualization, stress prediction, coefficient-based model explainability, and subject-wise generalization analysis, served by a frozen inference model.

HireSensei AI | AI-Powered Recruitment & Placement Platform | [GitHub Repo](#) | [Live Link](#)

Tech: React.js, TypeScript, Node.js, Express.js, MongoDB, Mongoose, JWT, REST APIs

- Architected a full-stack recruitment and placement platform connecting students, recruiters, companies, and colleges using React.js, TypeScript, Node.js, Express.js, MongoDB, and Mongoose.
- Implemented secure JWT authentication and role-based access control across Student, Recruiter, TPO, and Admin roles, with bcrypt password hashing, Zod validation, and protected REST APIs.
- Building the AI engineering layer with LLM-based resume analysis, RAG-based retrieval over placement and job knowledge, and ML-assisted candidate-job matching, integrated with the platform's structured recruitment and resume-processing pipeline.

Farm2Market | AI-Enabled Agri-Commerce Platform | [GitHub Repo](#) | [Live Link](#)

Tech: Python, TensorFlow, OpenCV, Machine Learning, Computer Vision, Full-Stack Web

- Engineered a full-stack agri-commerce marketplace integrating a machine learning-based fruit quality classifier and a data-driven price-prediction model to support fair, informed crop pricing.
- Built and integrated an image-analysis pipeline to assess produce freshness and grade, applying ML inference within a production-style full-stack application.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, Java, C, TypeScript

AI/ML: Machine Learning, Deep Learning, CNN, Computer Vision, Signal Processing, Time-Series Analysis, Feature Engineering, Explainable AI, Cross-Subject Validation, scikit-learn, TensorFlow, OpenCV, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Streamlit

Web & Backend: React.js, Node.js, Express.js, REST APIs

Databases: MongoDB, MySQL, PostgreSQL, Cassandra, Redis

Developer Tools: Git, GitHub, Linux, Docker, Postman, VS Code

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design

ACHIEVEMENTS AND CERTIFICATION

- Solved 200+ Data Structures & Algorithms problems, reflecting strong programming and problem-solving proficiency.
- Getting Started with Artificial Intelligence — IBM SkillsBuild

EDUCATION

B.Tech in Computer Science Engineering Kolhapur Institute of Technology CGPA: 7.6 / 10.0	2023 – 2027
12th Grade SRM Welkin Higher Secondary School, Sopore	83%
10th Grade SRM Welkin Higher Secondary School, Sopore	98%

LANGUAGES

English (Professional Proficiency) • Urdu/Hindi (Native or Bilingual Proficiency) • Kashmiri (Native Proficiency)